The Profit Leak Diagnostic

What a $35.2M Catalog's Revenue Report Doesn't Show You

Zeeshan Akram · E-commerce Profit Analyst

Public DataCo Supply Chain Dataset · 180,519 order lines · 3 years

The Verdict

This catalog generates $35.2 million in revenue. On paper, that looks like a healthy business. It isn't.

$35.2M	84.5%	28	100%	$643K
Total Revenue	**Profit From 8 Products**	**Low Performers**	**First Class Late Rate**	**Core 8 Stress Test**
180K+ orders, 3 years	Out of 118 total	Just 2.2% of revenue	Every single order	Modeled, 17% of profit

01 Every product in this catalog has recorded at least one loss-making order. There is no clean SKU in here.

02 Just 8 products generate 84.5% of all profit. The other 110 together earn about $590K.

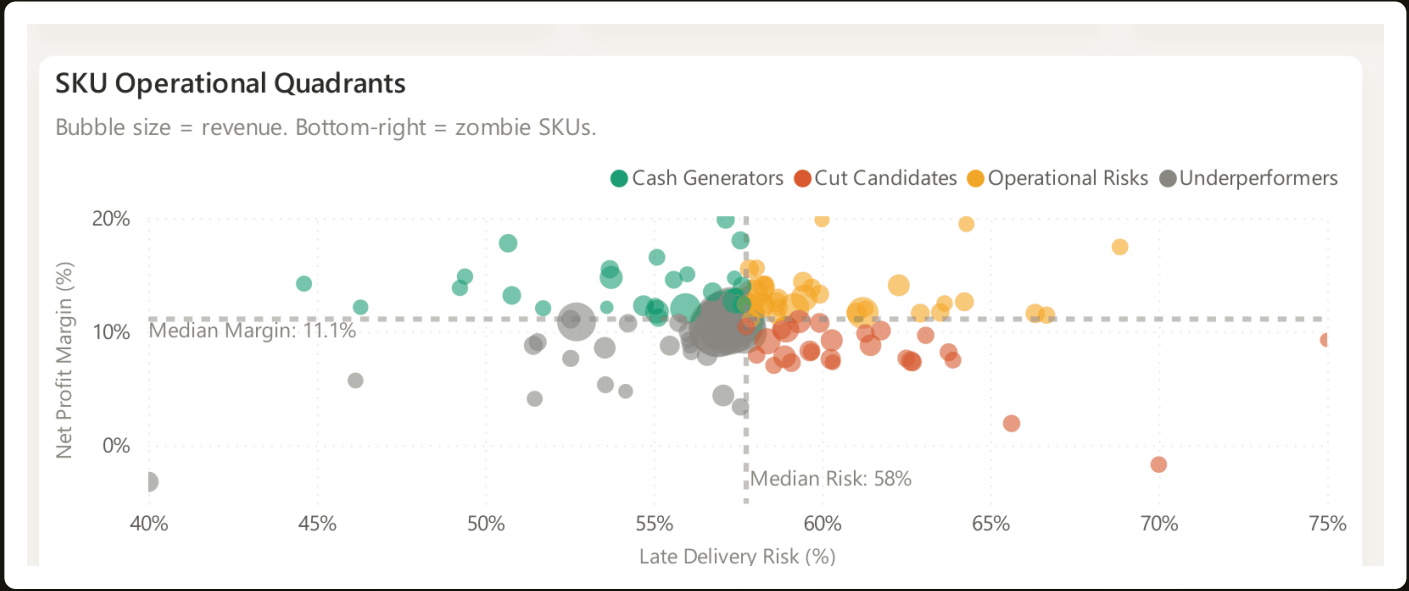
03 28 products generate only 2.2% of revenue and about $67K of profit. Sort them by cause before cutting any.

04 First Class shipping arrives late on every single order. Standard Class, the cheapest tier, is the most reliable shipping method in the catalog.

05 44 products sell far better than they earn. The SOLE E35 Elliptical ranks 36th by revenue and last by profit.

Revenue is not the issue here. Mistaking revenue for business health is.

Where Your Profit Actually Comes From



These boundaries are not arbitrary. They are set at the catalog's own median margin and median delivery risk, so every product is judged against its real peers.

Every product in the catalog is scored on two things that matter: how much margin it keeps, and how often it fails to deliver on time. That split sorts the whole catalog into four groups.

Cash Generators

Good margin, low risk. The products worth protecting.

Operational Risks

Decent margin, but failing on delivery often enough to threaten it.

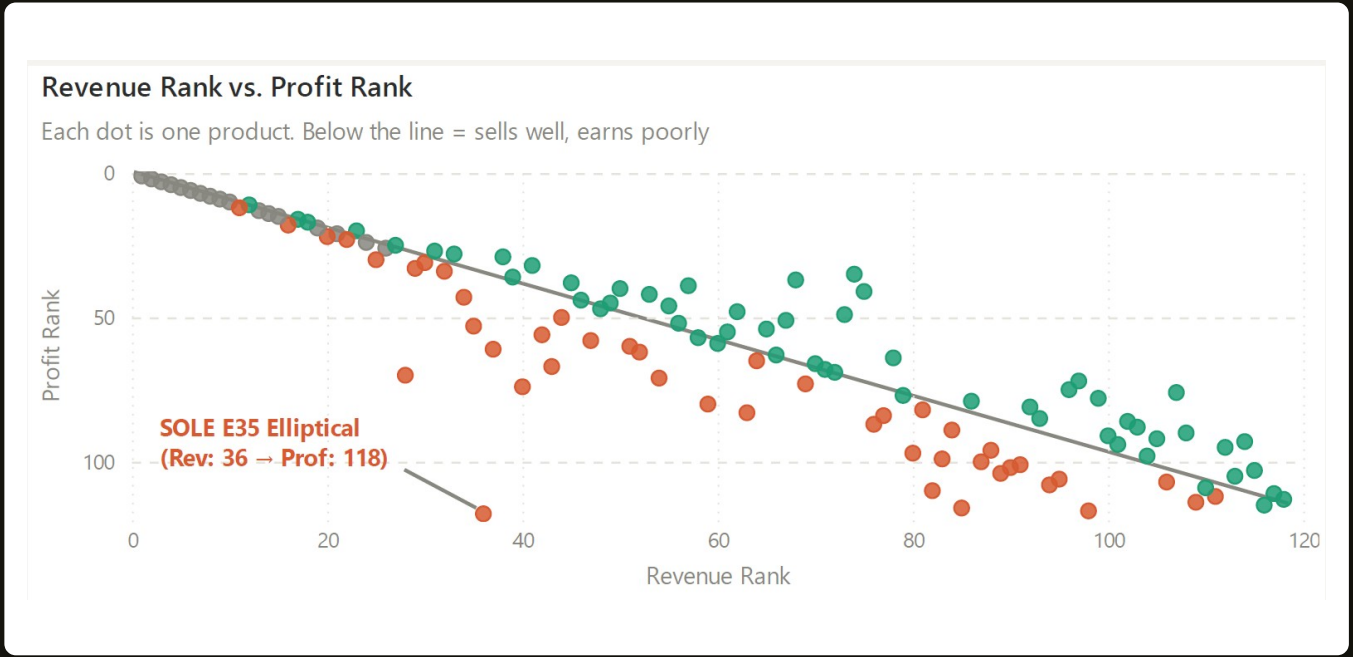
Underperformers

Margin already thin, even before anything goes wrong.

Cut Candidates

Weak margin and high delivery failure. 28 products, 24% of the catalog. Review by cause before cutting.

The Revenue Trap: Products That Sell Well and Lose Money



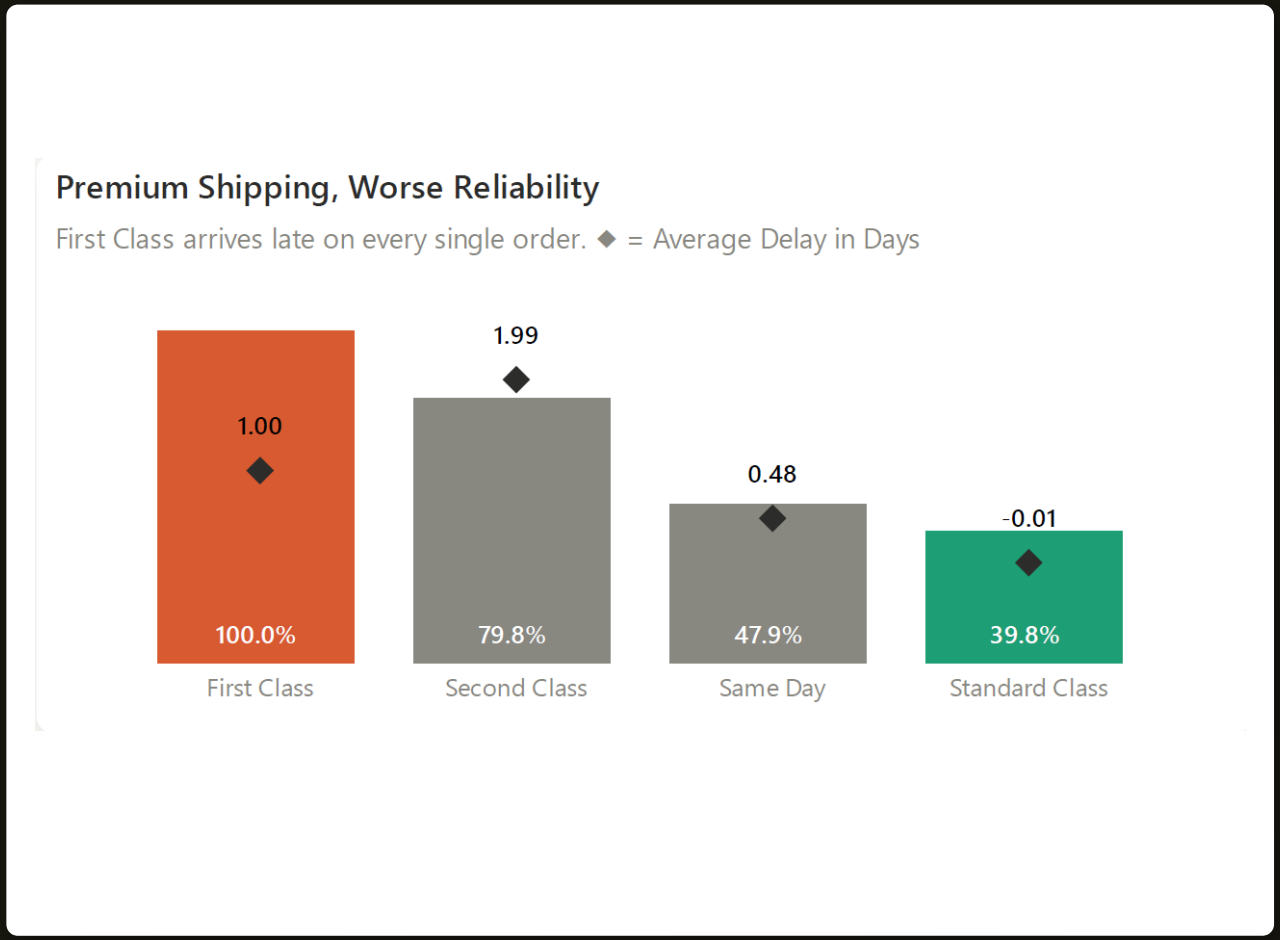
The SOLE E35 Elliptical ranks 36th out of 118 products by revenue. In profit, it ranks dead last. It is net negative over three years while showing up as a respectable mid-tier seller on any standard sales report.

This is not one unlucky product. 44 products in this catalog follow the same pattern: strong revenue, weak or negative profit.

44 products show this pattern. A revenue leaderboard ranks them higher than their profit deserves.

SOLE E35 and SOLE E25 Ellipticals are net negative across their entire three year history, not just in a bad month.

The Cost of Premium Shipping

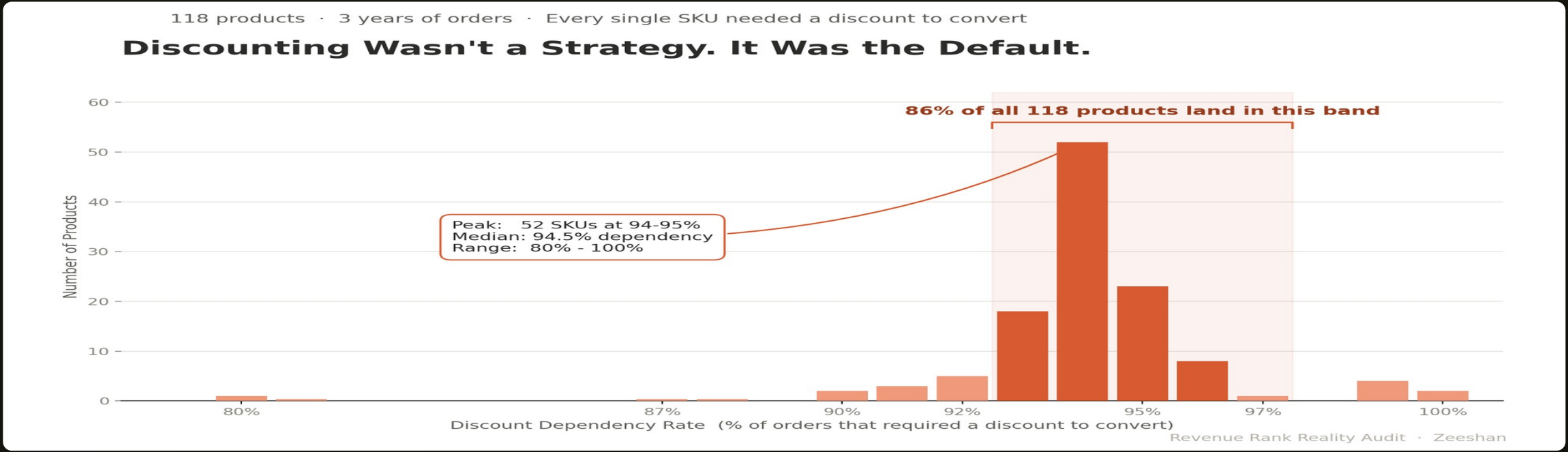


100% of First Class orders arrive late. Every order, every time.

Standard the cheapest tier available, is the most reliable shipping method in the entire catalog.

Customers paying the most for speed get the least reliable delivery. A 100% rate with a one-day average gap suggests the promised window may be mis-set. Check the delay distribution to confirm.

The Discounting Illusion



93–97%

of orders need a discount to sell, across almost the entire catalog

p = 0.73

no significant link between discount depth and margin

94.4%

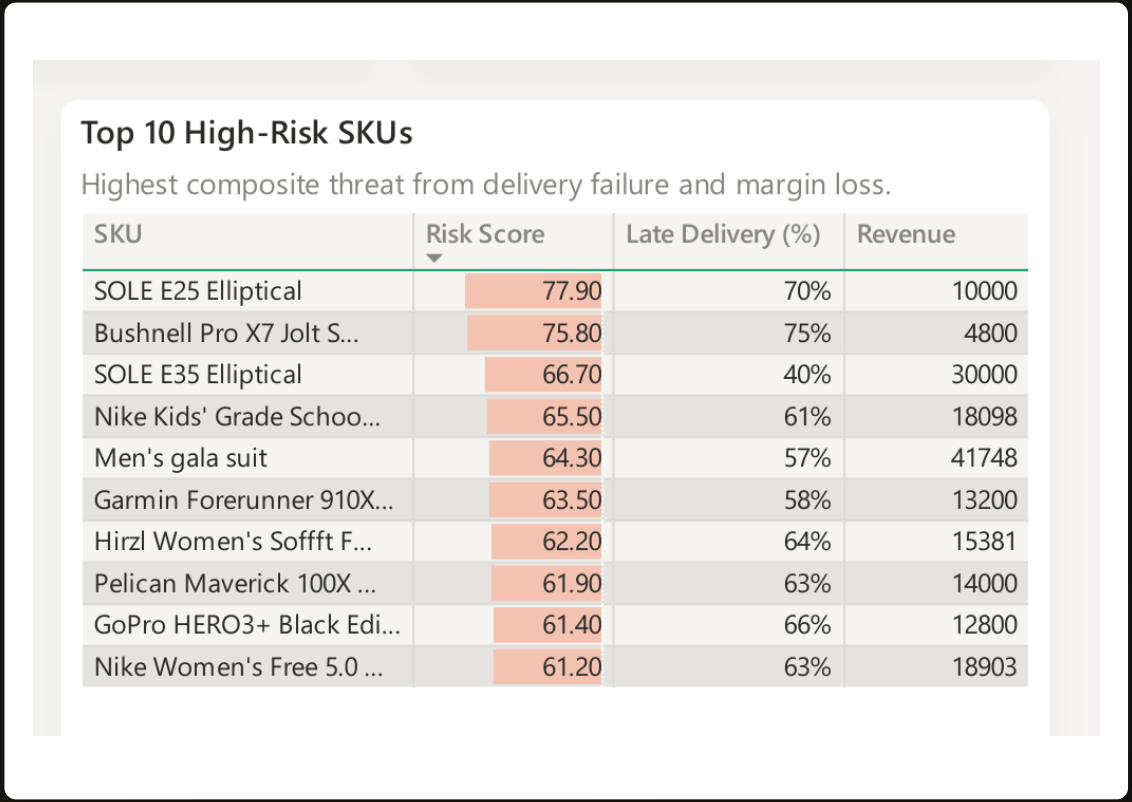
of orders from the top 4 products still require a discount

18.5%

of the top 4 products' orders still lose money

The top 4 earn the most in dollars, yet 94.4% of their orders carry a discount. Discount depth does not explain which products earn less.

The Risk Score: One Number Per Product



Top 10 High-Risk SKUs

Highest composite threat from delivery failure and margin loss.

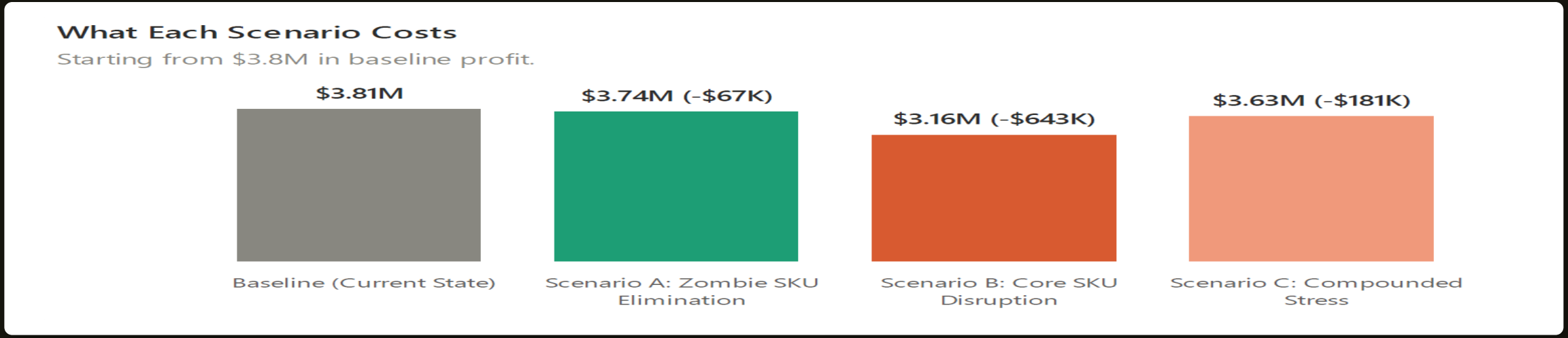
SKU	Risk Score	Late Delivery (%)	Revenue
SOLE E25 Elliptical	77.90	70%	10000
Bushnell Pro X7 Jolt S...	75.80	75%	4800
SOLE E35 Elliptical	66.70	40%	30000
Nike Kids' Grade Schoo...	65.50	61%	18098
Men's gala suit	64.30	57%	41748
Garmin Forerunner 910X...	63.50	58%	13200
Hirzl Women's Soffft F...	62.20	64%	15381
Pelican Maverick 100X ...	61.90	63%	14000
GoPro HERO3+ Black Edi...	61.40	66%	12800
Nike Women's Free 5.0 ...	61.20	63%	18903

Every product gets one combined score, built from three signals: how often it fails to deliver, how dependent it is on discounting, and how far its profit rank has drifted from its revenue rank.

On its own, none of these three signals reliably predicts which products are heading for trouble. Combined into one score, it does. The score is checked against actual outcomes across the catalog, not based on a guess about which SKUs look risky.

The value of this score is not retrospective. It flags a weakening product while it is still making sales, before the damage shows up as a quarter of disappointing revenue with no obvious cause.

What Each Scenario Costs



A DECISION

Retiring the 28 Low Performers

Gives up about $67,000 of profit (1.8%). Handling and support savings are not measured here, so review by cause first.

LARGEST EXPOSURE

A Disruption to the Core 8

A modeled 20% margin shock costs $643,000, more than the other 110 products earn combined.

DIRECTIONAL ESTIMATE

If Conditions Compound

+10% discount dependency and +15% delivery risk cost about $181,000. A directional estimate, not a forecast.

Data Rigor & Methodology

This audit is built on 180,519 order lines from the public DataCo dataset, spanning three years, loaded into a structured PostgreSQL warehouse and checked for data quality at every stage before a single chart was built.

1

Established in SQL

Every number above was first established in SQL as ground truth, before anything was charted.

2

Re-tested in Python

Each number was then independently reproduced and statistically tested in Python to confirm it holds up and is not just noise in the dataset.

3

Reported plainly either way

Findings were only treated as confirmed once they passed hypothesis testing at a 95% confidence threshold. Where a relationship did not hold, such as discount depth and margin, that absence was reported as plainly as the findings that did hold.

Nothing in this report is a guess dressed up as a conclusion.

What To Do Next

1. **Protect the core 8 products like they are the business.**

   Because they are. Lock in supplier terms, prioritize their fulfillment, and stop touching their pricing without a clear reason.

2. **Fix First Class, or reset its promise.**

   A tier that is late on every order is an active liability. Fix the process, or change the promised window to match what is delivered.

3. **Sort the 28 low performers by cause before cutting.**

   Retiring all 28 would give up about $67,000 of profit. Keep the new or seasonal, reprice the cost-driven, fix the delivery-driven, and retire only confirmed dead weight.

Let's Talk About Your Store

I run this exact diagnostic for e-commerce operators who need to know which SKUs are actually driving growth and which ones are quietly bleeding cash.

EMAIL

zeeshanakram1704@gmail.com

LINKEDIN

linkedin.com/in/zeeshan-akram-ds

BOOK A CALL

calendly.com/zeeshanakram/data-diagnostic